

DECODING THE HIDDEN RESERVOIR

A Forensic Approach to History Matching, Non-Uniqueness, and Probabilistic Forecasting

DEPTH: 12,450 FT

PRESSURE: 8,500 PSI

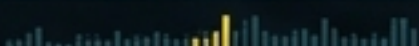
POROSITY: 18.5%

PERMEABILITY: 250 mD

DATA

DATA

DATA

DATA
INGESTIONMODEL
CALIBRATIONUNCERTAINTY
QUANTIFICATIONFORECAST
GENERATION

	INT1	INT2	DAT2	BEAG
N1	0.7A	0310	20	NA
N2	0.08	0010	20	NA
L3	0.7A	0102	00	NA

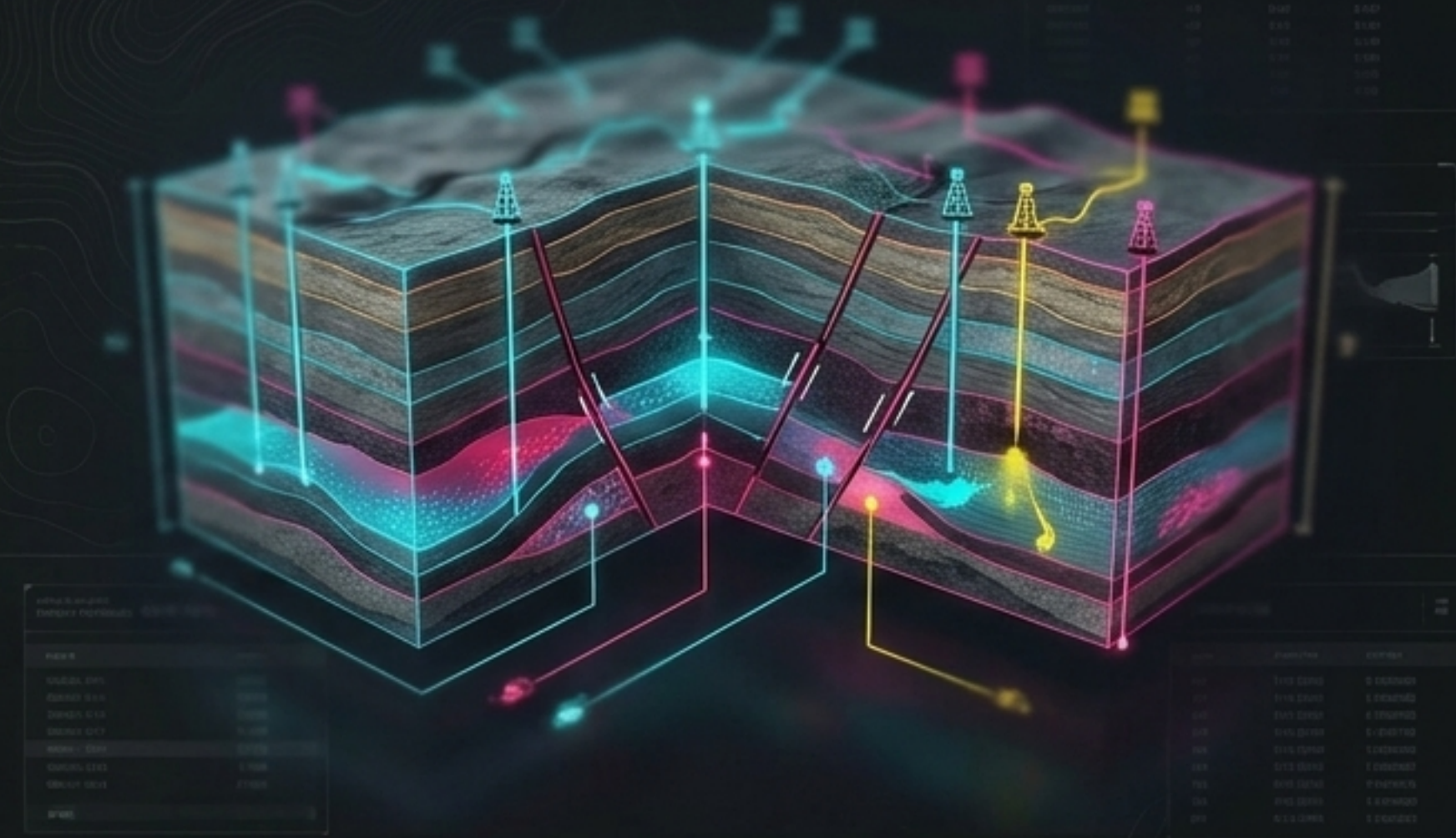


The Myth: Curve Fitting

- Adjusting random dials until the simulated line hits the historical data points.
- Treating the simulator as a black-box calculator.



The Paradigm: Forensic Decoding



- Adjusting reservoir model parameters to minimize the mismatch between observed historical data (oil, gas, water, pressures) and the simulator's physics engine.
- Viewing every mismatch as a clue to a hidden geological truth.

The Forensic Objectives



Validate Dynamics

Test the model's physics against reality: pressure depletion, water breakthrough, and gas coning.



Identify Heterogeneities

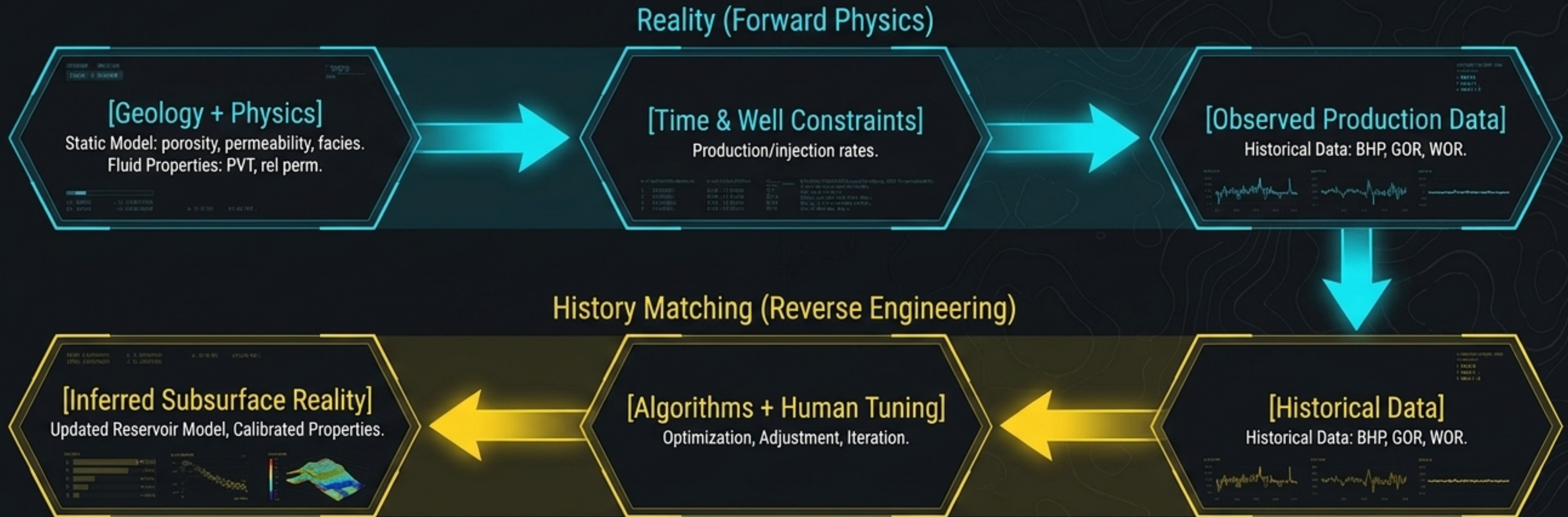
Locate the invisible barriers, unmapped faults, and key reservoir properties that static data alone cannot reveal.



Improve Confidence

Transform a theoretical static model into a reliable engine for field development and economic forecasting.

The Thinking Backwards Architecture



**We do not observe the reservoir; we observe its exhaust.
We must reverse-engineer the geology from the surface data.**

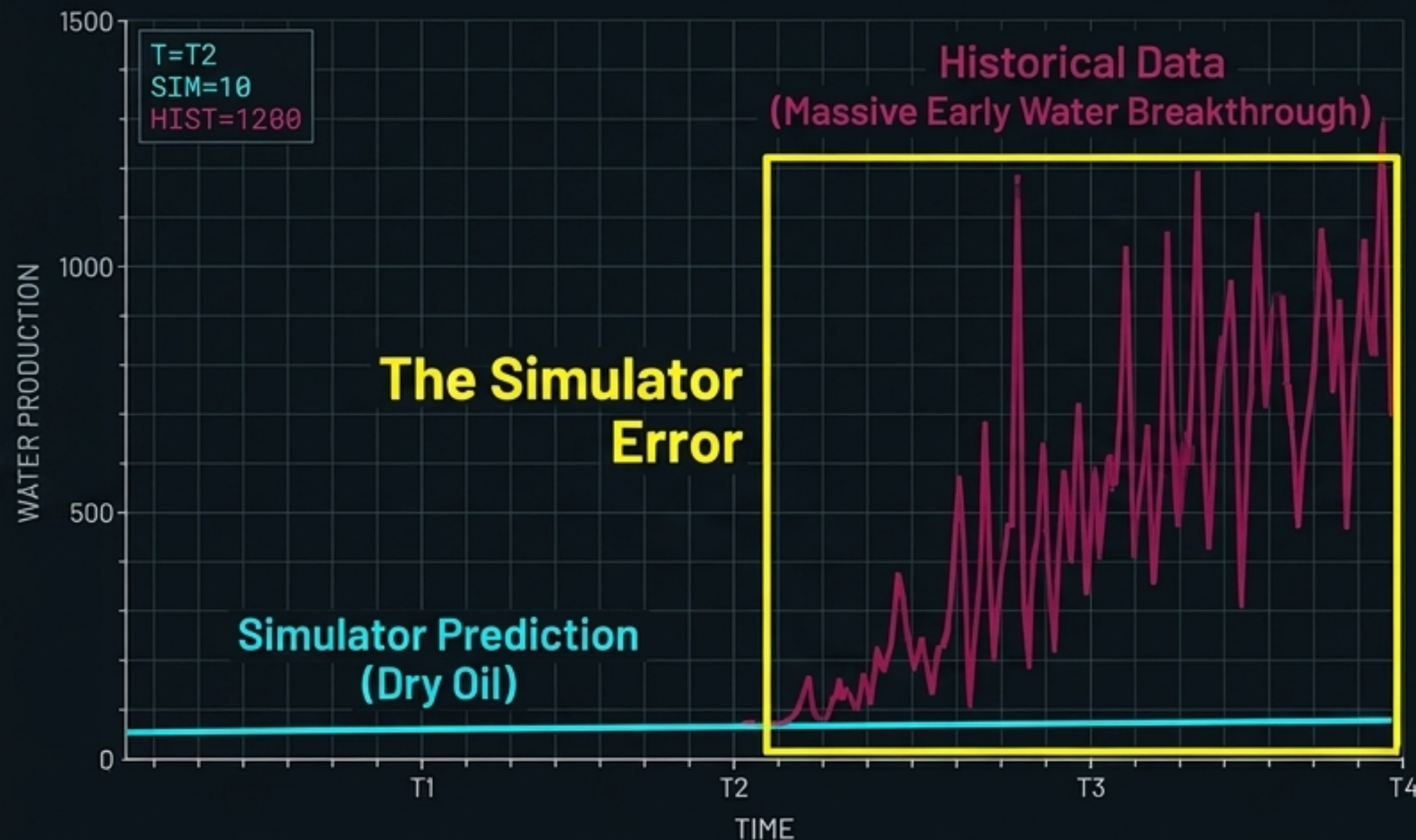
Mismatches Are the Reservoir Speaking

DATA SOURCE:

Eng.Sajjad k. Hano

T=T2, SIM=10, HIST=1200

HIST=0.00



INSIGHT

A perfectly matched curve teaches us nothing new. A severe mismatch reveals geological truths omitted from the static model.

Diagnostic Checklist:

- Unexpected aquifer strength?
- Unmapped high-permeability thief zones?
- Fault sealing behavior acting differently than assumed?

DATA SOURCE

TIME 05:59:00

TIME 20:53:39

NAME

COORDINATE -0.98

WATER -0.92

Interrogating the Suspects: The Parameter Bias Scale

Low Bias / Data-Driven

High Bias / Easily Manipulated

Anchor to Static Data

- Initial fluid contacts
- Rock properties (porosity-permeability)

Physics Tuning

- Relative permeability
- Capillary pressure curves

The 'Cheat' Dials

- Aquifer strength & boundaries
- Fault transmissibility & sealing behavior

Are you adjusting fault transmissibility because the geology supports it, or because it is the mathematically easiest way to force a match?

The Engines of Interrogation

Primary Engine

Manual History Matching

Assisted/Automated History Matching (AHM)

Engineering judgment & intuition

Optimization algorithms (Genetic, Particle Swarm, Gradient-based)

Iteration Method

Iterative tuning by trial and error

Evaluating thousands of model realizations based on an objective function (e.g., least squares error)

Bias Vulnerability

Highly susceptible to human confirmation bias

Blind to geology unless mathematically constrained

Resource Drain

Time-consuming and dependent on human experience

Computationally intensive for large-scale models requiring significant resources

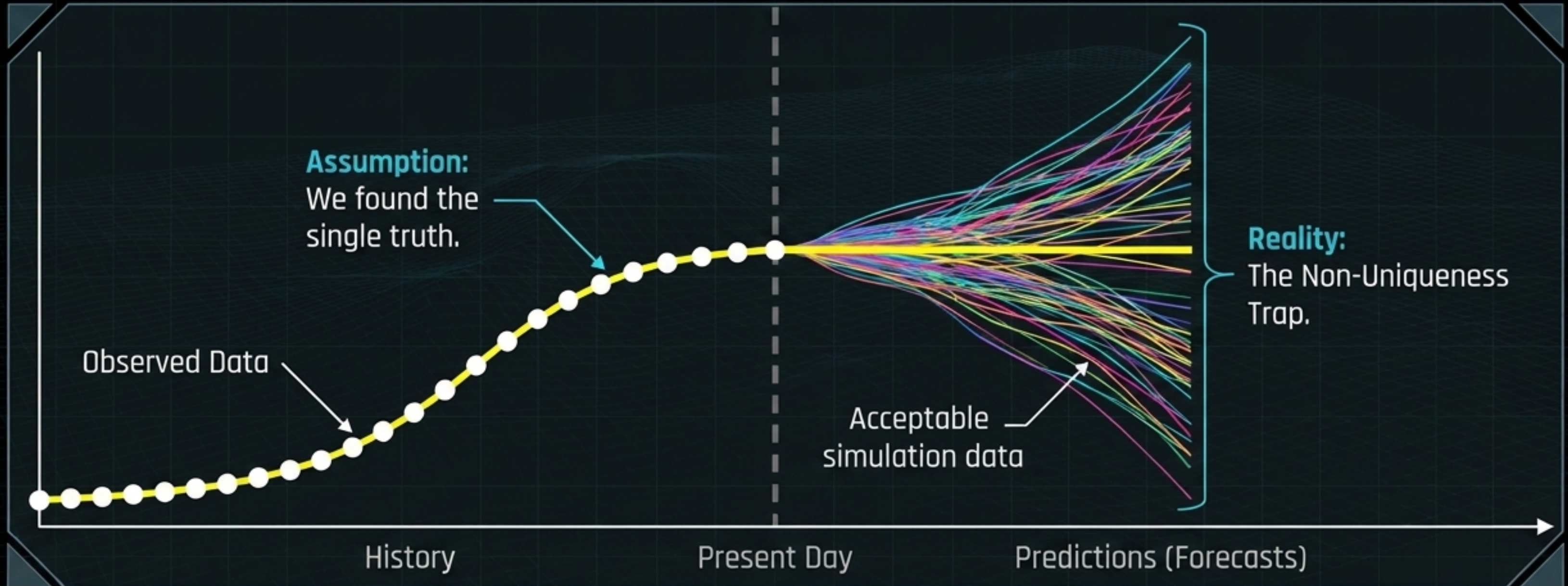


The Ultimate Trap: Non-Uniqueness

DEFINITION: Multiple parameter combinations may produce similar history matches.

Just because the simulator curve perfectly hits the historical data points does NOT mean your geological model is correct. A perfect match can be a mathematical illusion.

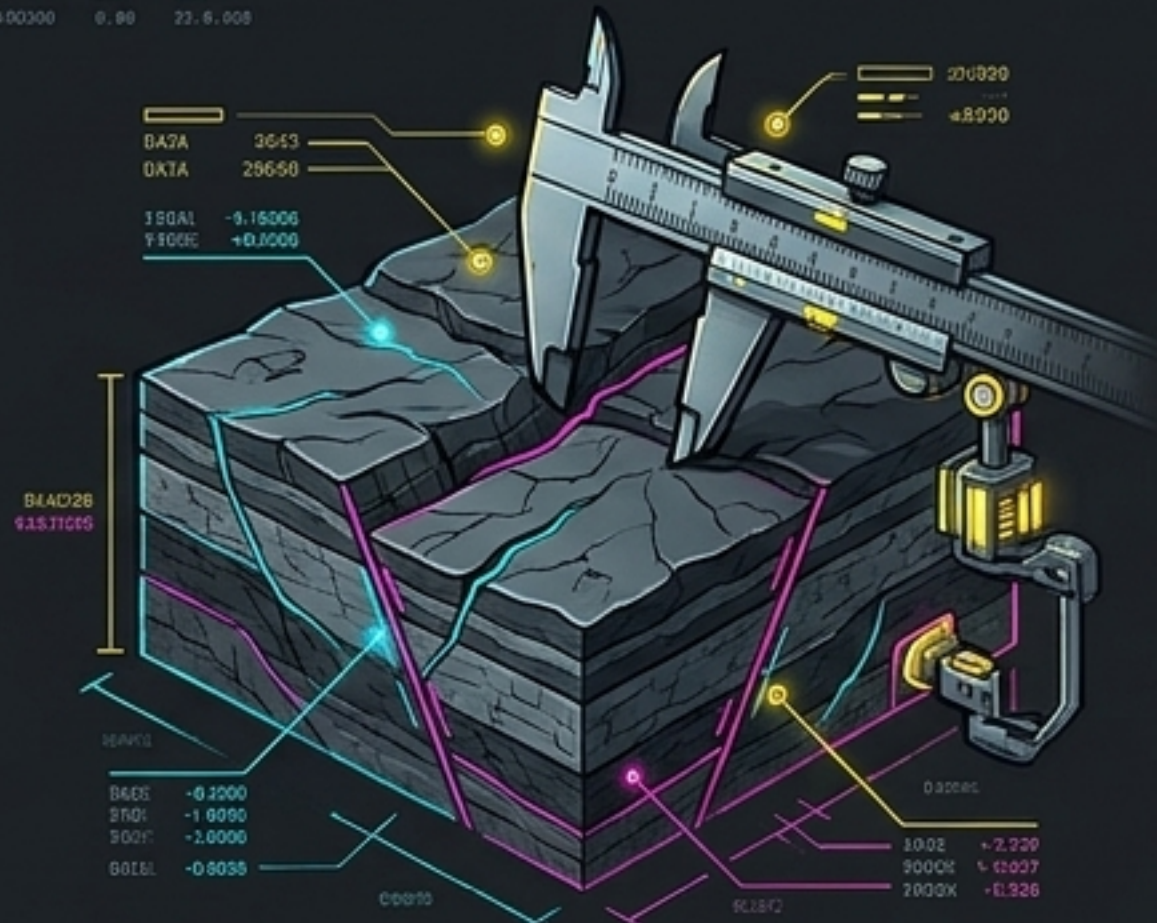
The Non-Uniqueness Mirage



Dozens of radically different geological models (faulted vs. homogeneous, strong vs. weak aquifers) can yield the exact same historical production curve, but wildly different forecasts.

Anchoring the Math to Reality

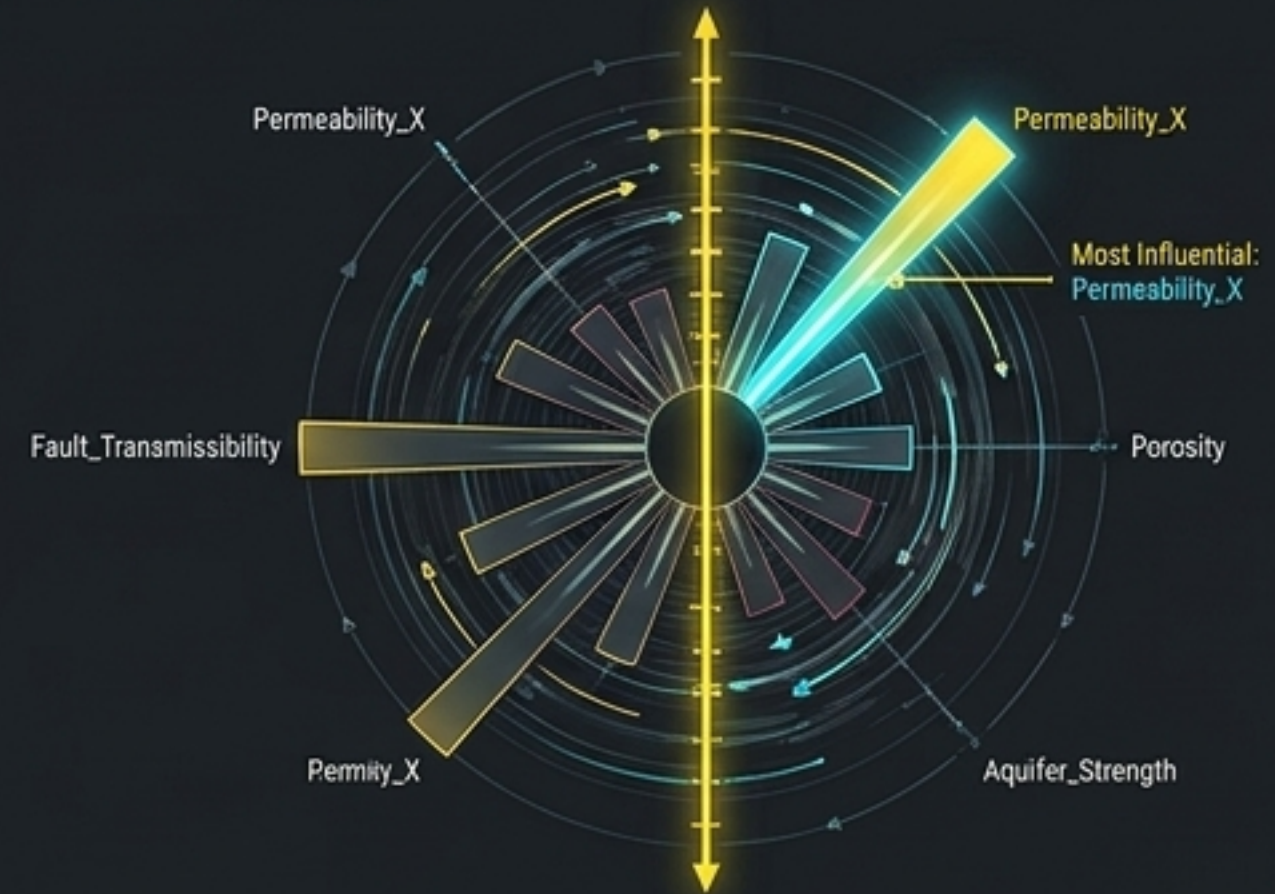
Geological Constraints



Do not let the optimization algorithms break the laws of geology.

Maintain realism by strictly bounding parameters (e.g., maximum possible fault throw, realistic facies distributions) while adjusting the model.

Sensitivity Analysis



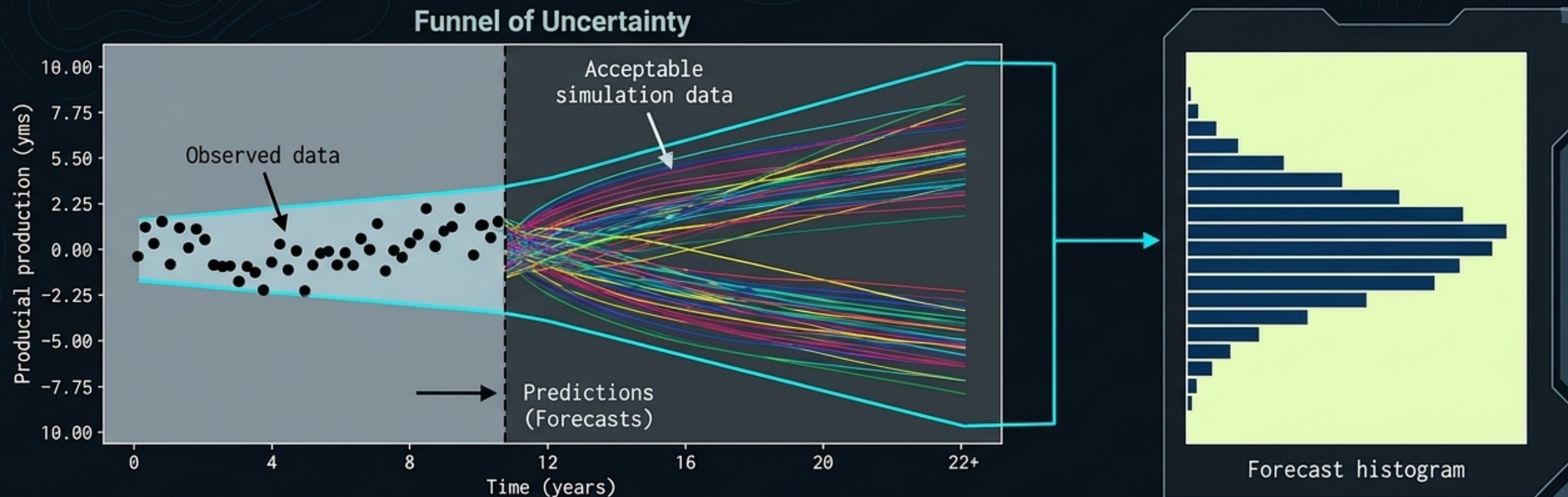
Before running thousands of realizations, identify which dials actually matter.

Apply rigorous sensitivity analysis to isolate the most influential parameters, reducing computational waste and noise.

Sequential Calibration & The Ultimate Test



THE VERDICT: EMBRACING PROBABILISTIC FORECASTING



Synthesis:

We must abandon the pursuit of one "perfect" deterministic model.

Core Mechanism:

By employing multiple realizations to capture uncertainty, we transform the non-uniqueness problem from a fatal flaw into a robust statistical forecast.

Final Insight:

History matching is not about eliminating uncertainty; it is about quantifying it to make highly informed, risk-adjusted decisions.